

Complete Revised Manuscript

Revised text, tables, and updated reference list

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Contents

1. Complete Revised Manuscript

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Supraorbital versus infraorbital external trigeminal nerve stimulation for migraine prevention: a three-arm assessor-blinded randomized sham-controlled trial

Abstract

Background

External trigeminal nerve stimulation is a non-invasive option for migraine prevention, but evidence directly comparing stimulation sites is limited. We evaluated supraorbital and infraorbital transcutaneous electrical nerve stimulation (TENS) against an inert sham condition.

Methods

This single-center, three-arm trial included 83 adults with migraine diagnosed according to International Classification of Headache Disorders, third edition criteria. Participants were allocated to supraorbital TENS (n=28), infraorbital TENS (n=27), or supraorbital sham stimulation (n=28). Active stimulation was administered in clinic at 100 Hz and 200 μ s for 30 min per session, six sessions weekly for four weeks. Sham stimulation used the same schedule but delivered no current. The primary outcome was monthly migraine attack frequency at three months. Secondary outcomes were pain intensity, Migraine Disability Assessment (MIDAS), Headache Impact Test-6 (HIT-6), acute oral analgesic-use frequency, and Migraine Prevention Questionnaire (MPQ-5) score. Outcome assessment and analysis were masked to allocation, but successful participant masking was not established.

Results

All 83 allocated participants were reported to have completed three-month follow-up. At three months, unadjusted mean monthly attack frequencies were 6.59 (SD 2.70), 6.16 (SD 2.48), and 12.94 (SD 5.01) in the supraorbital, infraorbital, and sham groups, respectively. The corresponding unadjusted endpoint mean differences versus sham were -6.35 attacks/month (approximate Welch 95% CI -8.52 to -4.18) and -6.78 attacks/month (-8.92 to -4.64). The supraorbital–infraorbital estimate was imprecise (0.43 attacks/month, -0.97 to 1.83) and does not establish equivalence. Descriptive active-versus-sham differences also favored active stimulation for pain intensity, MIDAS, HIT-6, and medication-related outcomes. Mild itching and tingling were common in both active groups. No serious adverse event was detected.

Conclusions

The aggregate data show a preliminary efficacy signal for both active protocols relative to an inert sham among patients able to attend an intensive clinic schedule. Unverified allocation and registration details, likely compromised participant masking, a site-mismatched sham, outcome-measure limitations, and insufficient power for the active-site comparison preclude confirmatory conclusions about efficacy or equivalence. Prospective-registration verification, source-data audit, and prespecified baseline-adjusted reanalysis are required before submission.

Trial registration: IRCT [AUTHOR TO INSERT IDENTIFIER, URL, AND REGISTRATION DATE].

Ethics approval: [AUTHOR TO INSERT COMMITTEE, APPROVAL IDENTIFIER, AND DATE].

First participant enrolled: [AUTHOR TO INSERT DATE].

Keywords: migraine; external trigeminal nerve stimulation; transcutaneous electrical nerve stimulation; supraorbital nerve; infraorbital nerve; randomized trial

1. Introduction

Migraine is a common neurological disorder and a leading cause of years lived with disability. The Global Burden of Disease 2023 analysis confirms that headache disorders remain highly prevalent and that migraine accounts for most headache-attributed disability [1]. Migraine is characterized by recurrent headache and associated sensory and autonomic symptoms, with diagnosis based on internationally standardized criteria [2]. An updated global synthesis estimated the prevalence of active migraine at 14.0% and headache on at least 15 days per month at 4.6% [3]. Migraine imposes substantial personal, economic, and health-system burdens, while underdiagnosis and undertreatment remain common [4].

Current pathophysiological models describe migraine as a disorder of sensory processing involving altered function across cortical, subcortical, brainstem, and trigeminovascular networks [5]. Preventive pharmacotherapy can reduce migraine burden, but contraindications, inadequate response, adverse effects, and medication overuse motivate evaluation of non-pharmacological options [6].

External trigeminal nerve stimulation delivers electrical stimulation through surface electrodes without an implanted device. Supraorbital and supratrochlear stimulation has been evaluated for migraine prevention in a small sham-controlled trial [7] and subsequent systematic reviews [8–11]. The evidence remains heterogeneous and device-specific. Current International Headache Society guidance provides conditional recommendations for named devices rather than treating all TENS devices, targets, and dose schedules as interchangeable [12].

Direct evidence for surface infraorbital stimulation in migraine is lacking. A small uncontrolled study evaluated combined supraorbital and infraorbital lidocaine blocks, but the contribution of the infraorbital component could not be isolated and nerve block cannot establish the efficacy of electrical stimulation [13]. A contemporary evidence review similarly identifies stronger evidence for supraorbital and occipital targets than for infraorbital stimulation in migraine [14].

We therefore evaluated supraorbital and infraorbital TENS against an inert supraorbital sham condition. The primary objective was to estimate the effect of each active treatment on monthly migraine attack frequency at three months relative to sham. The comparison between active sites was exploratory because the study was not designed for equivalence or non-inferiority.

2. Methods

2.1 Study design and oversight

This single-center, three-arm, parallel-group trial was conducted at the outpatient Physical Medicine and Rehabilitation clinics of Isfahan University of Medical Sciences, Iran, between [AUTHOR TO INSERT FIRST RECRUITMENT DATE] and [AUTHOR TO INSERT FINAL FOLLOW-UP DATE].

The protocol was approved by [AUTHOR TO INSERT FULL COMMITTEE NAME] on [DATE] (approval [IDENTIFIER]) and registered at the Iranian Registry of Clinical Trials on [DATE] (IRCT [IDENTIFIER]; [URL]). Written informed consent was obtained from all participants. The date of first participant enrollment was [DATE]. Any differences between the original registry/protocol and this report must be listed with their timing and rationale.

Reporting was revised with reference to CONSORT 2025, CONSORT Harms 2022, the CONSORT extension for non-pharmacological interventions, and TIDieR [15–18]. The dated protocol and statistical analysis plan should be made available at [REPOSITORY OR URL].

2.2 Participants

Adults aged 18–65 years were eligible if they had migraine with or without aura according to ICHD-3 criteria and at least two migraine attacks per month during the preceding three months [2]. A new attack was recorded after a pain-free interval of at least 24 hours. This definition should be reconciled with contemporary preventive-trial guidance and the exact diary algorithm should be supplied.

Exclusion criteria were use of preventive migraine medication within two weeks before enrollment; preventive procedures such as nerve block or botulinum toxin within the previous year; medication-overuse headache; another concurrent headache disorder; severe psychiatric or systemic illness; pregnancy or lactation; substance use disorder; and device-specific contraindications. Class-specific washout periods and any concomitant preventive treatment during follow-up must be reported because a universal two-week washout may be inadequate for long-acting treatments.

2.3 Recruitment and applicability

Of 137 screened individuals, 54 did not proceed to allocation because they were unable or unwilling to commit to 24 clinic visits over four weeks. The remaining 83 entered the allocation process. This high pre-allocation exclusion fraction limits applicability to patients able to attend an intensive clinic schedule. Because the 54 individuals were not randomized and had no study outcome observations, no outcomes were imputed for them.

2.4 Randomization and allocation concealment

The supplied report states that a computer-generated sequence and sequentially numbered, opaque, sealed envelopes were used. It also reports a fixed block size of four with a 1:1:1 allocation ratio. A conventional balanced 1:1:1 permuted block cannot have a size of four. Before submission, this section must be replaced using the original allocation records:

An independent [ROLE] generated the allocation sequence using [SOFTWARE, VERSION, AND EXACT ALGORITHM] with a 1:1:1 ratio and [ACTUAL BLOCK SIZES AND STRATIFICATION FACTORS]. [ROLE], who had no participant contact, prepared sequentially numbered, opaque, tamper-evident, sealed envelopes. After eligibility and consent were completed, [ROLE] opened the next envelope in sequence.

The original sequence, envelope log, and implementation roles require independent verification.

2.5 Masking

Participants were intended to be unaware of allocation. Active stimulation was titrated to a strong sensory level, whereas sham stimulation delivered no current. Successful participant masking therefore cannot be assumed. Outcome assessment and statistical analysis were conducted by an investigator unaware of allocation.

The original report states that 62% of participants in active groups and 54% in sham correctly guessed their allocation. A nonsignificant difference between these percentages does not establish successful masking. The final report should provide the complete assigned-group by guessed-group table, available response options, timing of assessment, reasons for guesses, and a suitable blinding index with confidence intervals. The design is described here as assessor-blinded.

2.6 Interventions

2.6.1 Supraorbital TENS

Bilateral stimulation was administered over the supraorbital foramina or notches using a TENS device manufactured by Novin Co., Iran [AUTHOR TO INSERT EXACT MODEL AND REGULATORY STATUS]. Sessions lasted 30 min and occurred six times weekly for four weeks, for 24 planned

sessions. Frequency was 100 Hz and pulse duration was 200 μ s. Current was increased to a strong but tolerable sensory level, targeting approximately 10 mA according to individual tolerance.

2.6.2 Infraorbital TENS

The same device, schedule, and nominal electrical parameters were used with bilateral electrodes centered over the infraorbital foramina.

2.6.3 Sham stimulation

Electrodes were placed over the supraorbital region using the same visible schedule and session duration, but no current was delivered. This inert sham did not reproduce treatment sensation and did not site-match the infraorbital intervention.

Before submission, the intervention description should additionally report waveform, polarity, electrode dimensions and material, spacing and montage, anatomical localization, skin preparation, impedance threshold, ramping, calibration, delivered-intensity distribution, provider training, protocol-fidelity checks, adherence, and deviations.

2.7 Outcomes

Outcomes were assessed at baseline, one month, and three months.

Primary outcome

The primary outcome was monthly migraine attack frequency at three months. The exact diary window, denominator, diary-completion threshold, and standardization method for incomplete or unequal windows must be supplied. Current preventive-trial guidance favors change in monthly migraine days from a prospective 4–8-week baseline [19]. Monthly migraine days and at least 50% responder rates should be reported if they were collected and prespecified.

Secondary outcomes

- 1. Pain intensity on a reported 0–10 scale. Authors must clarify whether this was a visual analogue scale or numerical rating scale, its anchors, recall window, and whether it represented average, worst, or attack-specific pain.
- 2. MIDAS total score using the validated Persian version [20]. MIDAS has a three-month recall period; therefore, a one-month measurement overlaps the pretreatment period and cannot represent a clean one-month treatment effect.
- 3. HIT-6 total score using the validated Persian version [21,22].
- 4. Number of acute oral analgesic-use occasions during the defined monthly window.
- 5. MPQ-5 score. The available citation is a meeting abstract and does not establish this instrument as a responsive continuous outcome [23]. Its range, scoring, Persian translation, licensing, and longitudinal validity require documentation. This endpoint should be treated as exploratory.

Adverse events were elicited by open-ended clinician inquiry at each visit. This approach may underdetect mild or unexpected events and should be acknowledged.

2.8 Concomitant treatment

Participants were permitted to use acetaminophen, non-steroidal anti-inflammatory drugs, and ergot derivatives for breakthrough attacks. The final report should provide medication classes and use by group and define whether new preventive treatments were prohibited, continued at a stable dose, or allowed during follow-up.

2.9 Sample size

The original protocol-based sample-size calculation must be reported exactly, including its endpoint, expected effect, variance or event rate, alpha, power, planned comparisons, test, and attrition allowance.

The supplied report retrospectively recalculated sample size for a continuous attack-frequency endpoint using an SD of 4.5, a three-attack difference, two-sided $\alpha=0.05$, and 80% power. A simple two-sample normal approximation under these assumptions requires approximately 36 evaluable participants per group. Allowing 20% attrition requires approximately 45 randomized participants per group after conservative rounding. This calculation does not account for two active-versus-sham comparisons and cannot replace the prospective design. Observed post hoc power has been removed.

2.10 Statistical analysis

The definitive analysis must be rerun from participant-level data under a dated analysis plan prepared without access to treatment-coded results. The recommended primary model is baseline-adjusted negative-binomial regression for monthly attack count, with log observation time as an offset if diary windows differ. The model should include treatment and baseline attack frequency, with two prespecified active-versus-sham contrasts and Holm or Dunnett control of the primary family-wise error rate. Adjusted rate ratios and absolute marginal mean differences should be reported with 95% confidence intervals.

Secondary outcomes should be analyzed using baseline-adjusted generalized models appropriate to their distributions. A longitudinal mixed model or generalized estimating equation including group, time, and group-by-time interaction may be used as a supportive analysis. The analysis population, missing-data strategy, and intercurrent-event strategy must be prespecified. The active-site contrast is exploratory and must not be interpreted as equivalence.

Because participant-level data were not provided for this editorial revision, the results below present reported group summaries and corrected approximate Welch endpoint confidence intervals. These aggregate estimates do not replace the required baseline-adjusted analysis. Separate within-group Friedman tests have not been used to establish efficacy.

3. Results

3.1 Participant flow

Of 137 screened individuals, 54 were excluded before allocation because they could not commit to the full session schedule. Eighty-three were allocated to supraorbital TENS (n=28), infraorbital TENS (n=27), or sham (n=28). The supplied report states that all allocated participants received the assigned intervention, completed the treatment schedule, and provided one- and three-month data. These statements should be verified against allocation, attendance, and diary logs and displayed in a CONSORT flow diagram.

3.2 Baseline characteristics

Participants had a mean age of approximately 35–36 years, and 63/83 (75.9%) were female. Table 1 presents the available demographic and outcome summaries. Statistical tests of baseline “homogeneity” were removed because baseline significance testing is not informative in a randomized trial and the reported age p value was not reproducible from aggregate data.

Table 1. Available baseline characteristics

Characteristic	Supraorbital (n=28)	Infraorbital (n=27)	Sham (n=28)
Age, years, mean (SD)	35.8 (7.6)	36.4 (6.9)	35.1 (7.2)
Female, n (%)	21 (75.0)	20 (74.1)	22 (78.6)
Migraine attacks/month	14.18 (3.95)	15.13 (4.51)	12.32 (4.99)
Pain intensity	7.30 (1.29)	7.07 (1.47)	7.10 (1.11)
MIDAS	20.02 (5.22)	17.94 (6.30)	19.36 (4.05)
HIT-6	62.58 (4.78)	64.22 (5.93)	61.11 (3.74)
Analgesic-use occasions/month	14.62 (4.35)	14.72 (3.50)	14.36 (4.27)
MPQ-5	26.43 (6.49)	26.52 (6.23)	25.42 (6.40)

The final table must add migraine with aura, chronic migraine, disease duration, headache days, migraine days, prior preventive failures, acute medication classes, comorbidities, and concurrent treatments.

3.3 Primary outcome

Reported monthly migraine attack frequencies are shown in Table 2. The sham group had a lower mean baseline frequency than both active groups. Unadjusted mean changes were –7.59, –8.97, and +0.62 attacks/month in the supraorbital, infraorbital, and sham groups, respectively.

Table 2. Monthly migraine attack frequency

Group	Baseline, mean (SD)	1 month, mean (SD)	3 months, mean (SD)	Unadjusted mean change
Supraorbital	14.18 (3.95)	7.68 (2.84)	6.59 (2.70)	−7.59
Infraorbital	15.13 (4.51)	9.19 (4.03)	6.16 (2.48)	−8.97
Sham	12.32 (4.99)	14.27 (4.15)	12.94 (5.01)	+0.62

At three months, the unadjusted endpoint mean differences were −6.35 attacks/month (approximate Welch 95% CI −8.52 to −4.18) for supraorbital TENS versus sham and −6.78 (−8.92 to −4.64) for infraorbital TENS versus sham. The supraorbital–infraorbital difference was 0.43 (−0.97 to 1.83). These estimates are descriptive and do not account for baseline differences.

The final manuscript must replace this paragraph with adjusted estimates from the audited participant-level analysis:

In the prespecified baseline-adjusted model, the estimated [RATE RATIO AND/OR ABSOLUTE MARGINAL DIFFERENCE] for supraorbital TENS versus sham was [ESTIMATE] (multiplicity-adjusted 95% CI [LOWER TO UPPER], p=[VALUE]), and for infraorbital TENS versus sham was [ESTIMATE] ([LOWER TO UPPER], p=[VALUE]). The exploratory supraorbital–infraorbital estimate was [ESTIMATE] ([LOWER TO UPPER]).

3.4 Pain intensity

Pain-intensity scores decreased numerically in both active groups and changed little in the sham group (Table 3).

Table 3. Pain-intensity scores

Group	Baseline, mean (SD)	1 month, mean (SD)	3 months, mean (SD)
Supraorbital	7.30 (1.29)	4.07 (0.76)	3.92 (0.97)
Infraorbital	7.07 (1.47)	5.05 (1.24)	3.90 (1.33)
Sham	7.10 (1.11)	6.16 (1.44)	6.53 (1.02)

At three months, descriptive differences versus sham were −2.61 (95% CI −3.14 to −2.08) for supraorbital TENS and −2.63 (−3.27 to −1.99) for infraorbital TENS.

3.5 Disability and headache impact

MIDAS and HIT-6 scores decreased numerically in both active groups (Tables 4 and 5). The one-month MIDAS values should not be interpreted as a pure one-month treatment response because the instrument uses a three-month recall period.

Table 4. MIDAS scores

Group	Baseline, mean (SD)	1 month, mean (SD)	3 months, mean (SD)
Supraorbital	20.02 (5.22)	12.33 (3.94)	9.76 (3.12)
Infraorbital	17.94 (6.30)	13.19 (4.62)	10.30 (2.07)
Sham	19.36 (4.05)	17.87 (5.20)	15.96 (5.39)

At three months, descriptive MIDAS differences versus sham were -6.20 (95% CI -8.57 to -3.83) and -5.66 (-7.88 to -3.44).

Table 5. HIT-6 scores

Group	Baseline, mean (SD)	1 month, mean (SD)	3 months, mean (SD)
Supraorbital	62.58 (4.78)	51.50 (3.61)	49.48 (4.84)
Infraorbital	64.22 (5.93)	51.50 (5.39)	51.40 (3.01)
Sham	61.11 (3.74)	62.39 (4.20)	62.18 (4.13)

At three months, descriptive HIT-6 differences versus sham were -12.70 (95% CI -15.11 to -10.29) and -10.78 (-12.73 to -8.83). A five-point within-person responder threshold is commonly used, but published estimates of clinically important change vary by method [24]. Clinical categories should be applied to individuals rather than group means.

3.6 Medication-related outcomes

Reported analgesic-use occasions and MPQ-5 scores decreased numerically in both active groups (Table 6). MPQ-5 is reported as exploratory because its longitudinal responsiveness and Persian validation were not documented.

Table 6. Medication-related outcomes

Outcome	Group	Baseline, mean (SD)	1 month, mean (SD)	3 months, mean (SD)
Analgesic-use occasions/month	Supraorbital	14.62 (4.35)	6.21 (2.59)	5.02 (2.64)
	Infraorbital	14.72 (3.50)	8.57 (3.20)	5.65 (2.55)
	Sham	14.36 (4.27)	12.72 (2.88)	12.06 (4.25)
MPQ-5	Supraorbital	26.43 (6.49)	16.73 (5.74)	16.77 (5.08)
	Infraorbital	26.52 (6.23)	15.98 (4.63)	14.87 (5.34)
	Sham	25.42 (6.40)	24.17 (6.49)	24.16 (4.23)

At three months, descriptive analgesic-use differences versus sham were -7.04 (95% CI -8.94 to -5.14) and -6.41 (-8.31 to -4.51). MPQ-5 differences were -7.39 (-9.90 to -4.88) and -9.29 (-11.91 to -6.67).

3.7 Summary of unadjusted three-month contrasts

Table 7. Descriptive endpoint comparisons

Outcome	Supraorbital–sham MD (95% CI)	Infraorbital–sham MD (95% CI)	Supraorbital–infraorbital MD (95% CI)
Migraine attacks/month	−6.35 (−8.52, −4.18)	−6.78 (−8.92, −4.64)	0.43 (−0.97, 1.83)
Pain intensity	−2.61 (−3.14, −2.08)	−2.63 (−3.27, −1.99)	0.02 (−0.61, 0.65)
MIDAS	−6.20 (−8.57, −3.83)	−5.66 (−7.88, −3.44)	−0.54 (−1.97, 0.89)
HIT-6	−12.70 (−15.11, −10.29)	−10.78 (−12.73, −8.83)	−1.92 (−4.10, 0.26)
Analgesic-use occasions	−7.04 (−8.94, −5.14)	−6.41 (−8.31, −4.51)	−0.63 (−2.03, 0.77)
MPQ-5	−7.39 (−9.90, −4.88)	−9.29 (−11.91, −6.67)	1.90 (−0.92, 4.72)

The confidence intervals in Table 7 are approximate Welch mean-difference intervals reconstructed from aggregate summaries. They do not correspond to rank-based estimands and must be replaced by adjusted model estimates after source-data reanalysis.

3.8 Adherence and treatment exposure

The supplied report states that all randomized participants completed all 24 sessions. The final manuscript should report sessions received as median (IQR), the number completing all sessions, actual delivered-current distribution, protocol deviations, and diary completeness by group.

3.9 Adverse events

Itching was reported by 9/28 (32.1%) participants in the supraorbital group and 8/27 (29.6%) in the infraorbital group. Tingling or paresthesia was reported by 12/28 (42.9%) and 11/27 (40.7%), respectively. Symptoms were described as transient and did not lead to discontinuation. No serious adverse event was detected in any group.

Sham-arm local-event counts, event severity, duration, relatedness, repeated occurrences, and actions taken must be reported. Zero serious events in 83 participants does not establish safety; the approximate rule-of-three upper 95% bound for an unobserved serious-event proportion is 3.6% overall.

4. Discussion

4.1 Principal findings

The reported aggregate data showed lower migraine attack frequency at three months in both active-stimulation groups than in the inert sham group. Similar descriptive patterns occurred for pain intensity, disability, headache impact, and medication-related measures. These observations

constitute a preliminary efficacy signal rather than confirmatory evidence because the primary analysis requires baseline adjustment, participant masking was not established, and the sham condition did not reproduce treatment sensation.

The active-site contrast was imprecise and the study was not designed for equivalence or non-inferiority. Absence of a statistically significant difference cannot establish that supraorbital and infraorbital stimulation are equivalent, interchangeable, or similarly effective.

4.2 Relation to previous evidence

The pivotal PREMICE trial evaluated a proprietary supraorbital/supratrochlear device using 60 Hz, 250 μ s, up to 16 mA, and 20 min daily for three months [7]. The present protocol used a different device, target montage, 100 Hz, 200 μ s, and 30 min six times weekly for four weeks. Results from one protocol cannot be transferred directly to another.

Systematic reviews have reported possible preventive benefits from external trigeminal or transcutaneous nerve stimulation, but they included few heterogeneous studies and generally called for larger, higher-quality trials [8–11]. A broader meta-analysis of eTNS across neurological and psychiatric indications found no significant benefit for several migraine monotherapy outcomes, illustrating uncertainty in the evidence base [25]. Current IHS guidance therefore makes conditional, device-specific recommendations and does not endorse generic TENS or infraorbital stimulation as a class [12].

A 2026 open-label active-controlled trial compared supraorbital tSNS with low-dose topiramate [26]. The tSNS group had fewer adverse events but a lower responder rate (20.6% vs 33.9%) and did not meet the prespecified non-inferiority criterion. This recent evidence provides useful comparative context but does not validate the present protocol because the device, comparator, masking, and schedule differed.

The TEAM trial provided stronger evidence for supraorbital eTNS in acute treatment, but it evaluated a single acute attack rather than preventive treatment [27]. Acute and preventive claims require different trial designs and should not be conflated.

4.3 Infraorbital stimulation and possible mechanisms

Direct randomized evidence for surface infraorbital stimulation in migraine remains absent. The combined nerve-block study [13] cannot isolate the infraorbital contribution, and implanted peripheral-stimulation evidence largely concerns other targets or neuropathic craniofacial pain [14].

V1 and V2 afferents have distinct peripheral anatomy but converge within central trigeminal pathways [28]. Proposed mechanisms of external trigeminal stimulation include modulation of trigeminal nociceptive transmission, central pain-processing networks, autonomic signaling, and neurotransmitter or neuropeptide pathways [29]. These mechanisms are largely indirect, preclinical,

or derived from supraorbital stimulation. The numerical one-month difference between active sites was not tested with a group-by-time interaction and should not be interpreted as evidence of faster supraorbital action.

4.4 Clinical interpretation

The intervention required 24 supervised clinic sessions in four weeks, and 54/137 screened individuals did not proceed because they could not commit to that schedule. Effectiveness, feasibility, cost, and adherence in routine care may therefore differ from outcomes in the selected sample.

The inert sham produced no stimulation sensation, whereas active treatment was titrated to strong paresthesia. Because efficacy outcomes were participant-reported, expectation and performance effects may account for part of the observed active–sham difference. The site mismatch also means that the infraorbital-versus-sham comparison does not isolate stimulation from placement-related expectations.

The study does not establish long-term durability, broad safety, acute abortive efficacy, or comparative effectiveness against established preventive options. Contemporary recommendations recognize selected non-invasive neuromodulation devices as possible options for some patients, but such guidance cannot be generalized to an unvalidated device, montage, or target [12,30].

4.5 Strengths and limitations

Strengths include a three-arm controlled design, direct evaluation of two trigeminal sites, multidimensional outcomes, blinded outcome assessment and analysis, and reported complete follow-up after allocation.

The study has substantial limitations:

- 1. Ethics approval and prospective registration cannot be verified until identifiers, dates, and records are supplied.
- 2. A fixed block size of four is incompatible with a conventional balanced 1:1:1 permuted-block design and requires source verification.
- 3. The original sample-size basis and timing of changes to the outcome or analysis are unclear.
- 4. The reported analysis did not adjust for baseline and relied heavily on separate within-group tests.
- 5. The inert sham likely compromised sensory masking, and all efficacy outcomes were participant-reported.
- 6. Sham placement matched only the supraorbital site.
- 7. The study was not powered for equivalence or non-inferiority between active sites.
- 8. Exclusion of 54 screened individuals based on ability to attend restricts applicability.
- 9. Attack definition, diary windows, diary completeness, and derivation of non-integer monthly frequencies require clarification.
- 10. MIDAS at one month overlaps the pretreatment recall period; pain-scale and MPQ-5 measurement properties were incompletely specified.

- 11. Clinical baseline characteristics and concomitant-treatment data were insufficient.
- 12. Open-ended adverse-event ascertainment may have underdetected harms.
- 13. The single-center sample was small and predominantly female, with only three months of follow-up.

The previously reported analysis assigning nonrandomized individuals to sham was invalid and has been removed. It could not address transportability and was numerically inconsistent with the reported tables.

4.6 Future research

A definitive study should follow IHS neuromodulation-device trial guidance [19] and use a verifiable prospectively registered protocol and statistical analysis plan; concealed allocation with valid variable block sizes; an active sensory sham matched to each electrode site; blinded outcome assessment; standard 28-day migraine-day outcomes; validated patient-reported measures; class-specific medication rules; adequate power for intended contrasts; systematic harms capture; and longer follow-up. A factorial or double-dummy design could separate stimulation-site effects from sham-site and sensory-expectation effects.

5. Conclusions

Among a selected group able to attend 24 clinic sessions, the reported aggregate data showed fewer self-reported migraine attacks at three months with supraorbital and infraorbital TENS than with an inert sham. These findings are hypothesis-generating. Unverified allocation and registration details, likely compromised participant masking, a site-mismatched sham, outcome and analysis limitations, and inadequate power for the active-site comparison preclude firm conclusions about efficacy, safety, or equivalence. Source-document verification and prespecified participant-level reanalysis are necessary before the findings can support clinical recommendations or journal submission.

Acknowledgments

The authors thank the participants and clinical staff of the Physical Medicine and Rehabilitation clinics at Isfahan University of Medical Sciences.

Ethics approval and consent to participate

[AUTHOR TO INSERT FULL ETHICS COMMITTEE, APPROVAL IDENTIFIER, AND DATE.] All participants provided written informed consent.

Trial registration

Iranian Registry of Clinical Trials: [AUTHOR TO INSERT IRCT IDENTIFIER, URL, REGISTRATION DATE, AND FIRST ENROLLMENT DATE].

Funding

The supplied report states that this research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Authors must verify this declaration.

Competing interests

The authors declare [AUTHOR TO INSERT VERIFIED DECLARATION].

Data and code availability

De-identified participant-level data, the data dictionary, and executable analysis code will be available at [AUTHOR TO INSERT REPOSITORY], subject to applicable ethics and privacy requirements.

Author contributions

[AUTHOR TO INSERT COMPLETE CRediT ROLES FOR EACH NAMED AUTHOR.]

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